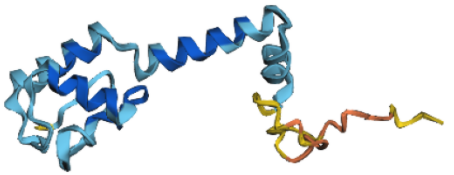
Four systems on one held-out specification (A6LEG8)

Prompt: 126 residues · 1 cysteine · stable · low hydrophobicity · no transmembrane helix · PF00416. All four structures share one camera and one origin, so lengths are directly comparable; chains run N→C. Colour is per-residue ESMFold confidence. Self-play is the only arm that fails, and the only one above pLDDT 90.

Natural A6LEG8

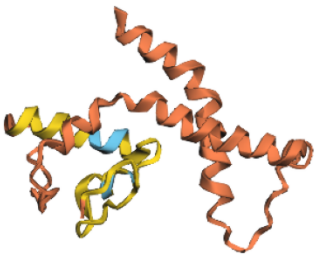
pLDDT 78.8
126 res · max AA 0.135
3.89 bits



SFT

pLDDT 43.1
137 res · max AA 0.161
3.82 bits

passes gate + filters



Self-play DPO

pLDDT 93.8
194 res · max AA 0.351
2.88 bits

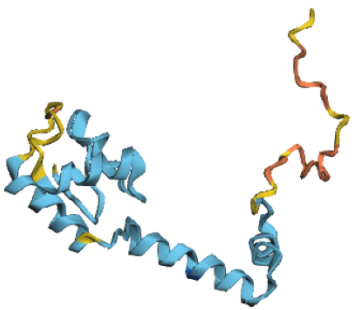
FAILS gate + filters



Anchored DPO

pLDDT 71.7
124 res · max AA 0.153
3.83 bits

passes gate + filters



50 Å

pLDDT > 90 · 70–90 · 50–70 · < 50